

Project 1 Report

Title: Fuzzy Time Series Forecasting (FOFTS & HOFTS)

Course: Fuzzy System

Executive Summary

This project presents a fully from-scratch implementation of a Fuzzy Time Series (FTS) forecasting framework, including both First-Order FTS (FOFTS, order = 1) and High-Order FTS (HOFTS, order ≥ 2). The entire pipeline—covering universe of discourse construction, fuzzy partitioning, membership function design, fuzzification, FLRG generation, defuzzification, and forecasting—was implemented without relying on any pre-built fuzzy time series libraries, in accordance with the assignment requirements.

A systematic grid search was performed over model order and number of partitions, and all configurations were evaluated using RMSE, MAE, and MAPE on a time-ordered 80%/20% train-test split with one-step-ahead prediction and teacher forcing.

The best-performing configurations selected based on RMSE were:

- **Mackey–Glass:** order = 3, partitions = 30, RMSE = 0.034076, MAE = 0.025283, MAPE = 4.075%.
- **Influenza — TOTAL SPECIMENS:** order = 1, partitions = 15, RMSE = 10617.999.
- **Influenza — TOTAL A:** order = 3, partitions = 30, RMSE = 3700.406 (log1p scale during training).
- **Influenza — TOTAL B:** order = 1, partitions = 15, RMSE = 332.418 (log1p scale during training).

For the Mackey–Glass dataset, the best HOFTS configuration achieved an RMSE reduction of approximately 32% compared to the best FOFTS baseline, demonstrating the ability of higher-order fuzzy models to capture long-range temporal dependencies in chaotic systems. In contrast, for the Influenza datasets, higher-order models provided only marginal improvements for TOTAL A and no

improvement for TOTAL B, which can be attributed to increased rule sparsity and the inherently noisy nature of epidemiological count data.

Overall, the results confirm that HOFTS models are particularly beneficial for deterministic nonlinear systems, while FOFTS remains competitive for sparse or highly stochastic real-world data.

Methodology

Data preprocessing and temporal split

Each dataset is treated as a univariate time series and processed independently. All missing, non-numeric, or invalid observations are removed prior to modeling. To preserve the temporal structure of the data, a strictly time-ordered split is applied, where the first 80% of observations are used for training and the remaining 20% for testing. No random shuffling is performed, as this would violate the forecasting setting.

Model evaluation is conducted using one-step-ahead prediction with teacher forcing. At each test step, the true observation is appended to the historical sequence before predicting the next value. This strategy ensures that prediction errors do not accumulate over time and allows fair comparison across different model configurations.

Universe of discourse and fuzzy partitioning

For each training series, the universe of discourse U is defined as:

$$U = [\min(X_{train}) - \delta, \max(X_{train}) + \delta]$$

where $\delta = 0.05 \times (\max(X_{train}) - \min(X_{train}))$ is a margin introduced to avoid boundary effects.

The universe is partitioned into m fuzzy sets, where $m \in \{7, 10, 15, 20, 30\}$. Two partitioning strategies are implemented:

1. **Equal-width partitioning:** fuzzy set centers are uniformly spaced across the universe.

2. **Quantile-based partitioning:** fuzzy set centers are placed according to empirical quantiles of the training data, allocating more resolution to dense regions.

For the Influenza datasets, both strategies are evaluated and the best one is selected based on RMSE.

Membership functions and fuzzification

Both triangular and Gaussian membership functions are implemented in the framework. In all reported experiments, triangular membership functions are used due to their simplicity, interpretability, and low computational cost.

Given a crisp value x , its membership degrees are computed for all fuzzy sets A_i . Fuzzification is performed using the maximum membership principle:

$$f(x) = \arg \max_i \mu_{A_i}(x)$$

This produces a discrete fuzzy state sequence $\{A_1, A_2, \dots\}$, which serves as the symbolic representation of the original time series.

FLRG generation and forecasting

Fuzzy Logical Relationships (FLRs) are constructed from the fuzzified training sequence. For FOFTS (order = 1), relationships take the form:

$$A(t-1) \rightarrow A(t)$$

For HOFTS (order = K), relationships are defined as:

$$(A(t-K), \dots, A(t-1)) \rightarrow A(t)$$

All FLRs are stored as frequency-aware rule groups (FLRGs), implemented as mappings from antecedent patterns to counters of consequent states. This representation preserves information about the empirical distribution of outcomes.

During forecasting, defuzzification is performed using the weighted average of consequent centers:

$$\hat{y}(t) = \frac{\sum_j c_j \cdot w_j}{\sum_j w_j}$$

where c_j is the center of fuzzy set A_j and w_j is its frequency in the matched FLRG. If no rule is found, a persistence fallback strategy is used, returning the last observed value.

True back-off strategy for HOFTS

To mitigate the severe sparsity problem inherent in high-order fuzzy models, a true back-off mechanism is implemented. During training, separate FLRG tables are constructed for all orders from 1 up to K .

During inference, the model first attempts an exact match in the order- K table. If no rule is found, it progressively backs off to lower-order tables by matching shorter suffix patterns, until a valid rule is obtained (down to order 1). This strategy significantly reduces fallback events and increases effective rule coverage.

Metrics and experimental protocol

For each configuration of (order, number of partitions), model performance is evaluated on the test segment using RMSE, MAE, and MAPE. The final model selection criterion is RMSE.

For Influenza A and B datasets, a log1p transformation is applied during training to stabilize variance and reduce skewness. Predictions are inverted back to the original scale using the exponential function and clamped to non-negative values. All reported metrics for these targets are computed on the original scale.

Results and Analysis

Dataset 1 — Mackey–Glass

Model	order	partitions	RMSE	MAE	MAP E	n_rules_orderK	hit_rate_any	fallback_rate	backoff_rate
Best (selected by RMSE)	3	30	0.0340761	0.0252832	4.07495	232	0.97	0.03	0.09
Best FOFTS (order=1)	1	30	0.0501102	0.0403022	6.76605	28	0.97	0.03	0

The best Mackey–Glass configuration is order=3 with 30 partitions. The model maintains a high rule-hit rate due to the back-off mechanism and a low fallback rate, indicating that the learned FLRGs cover most test patterns.

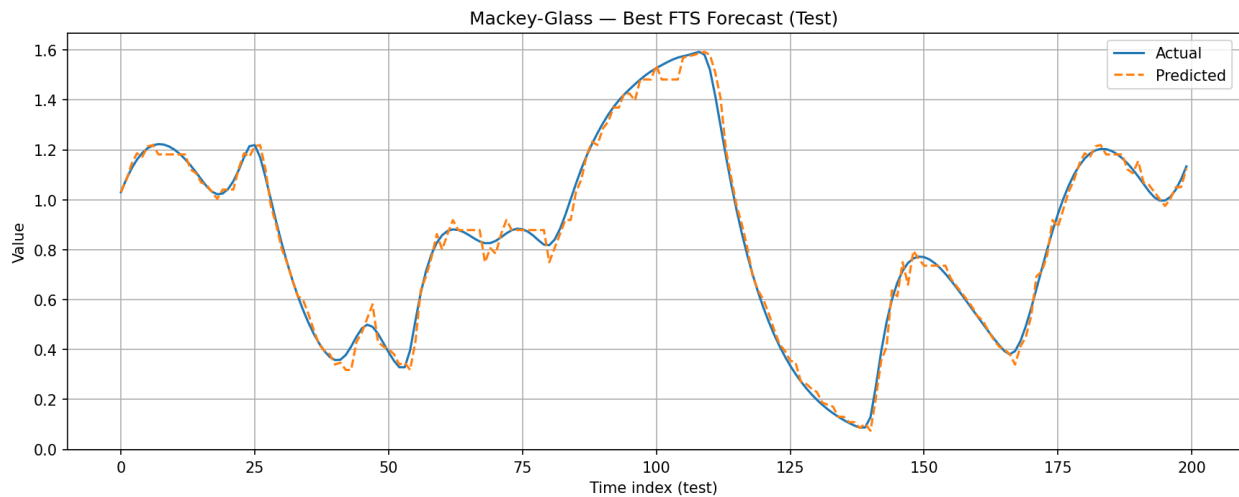


Figure 1. Mackey–Glass: actual vs predicted (best configuration, test segment).

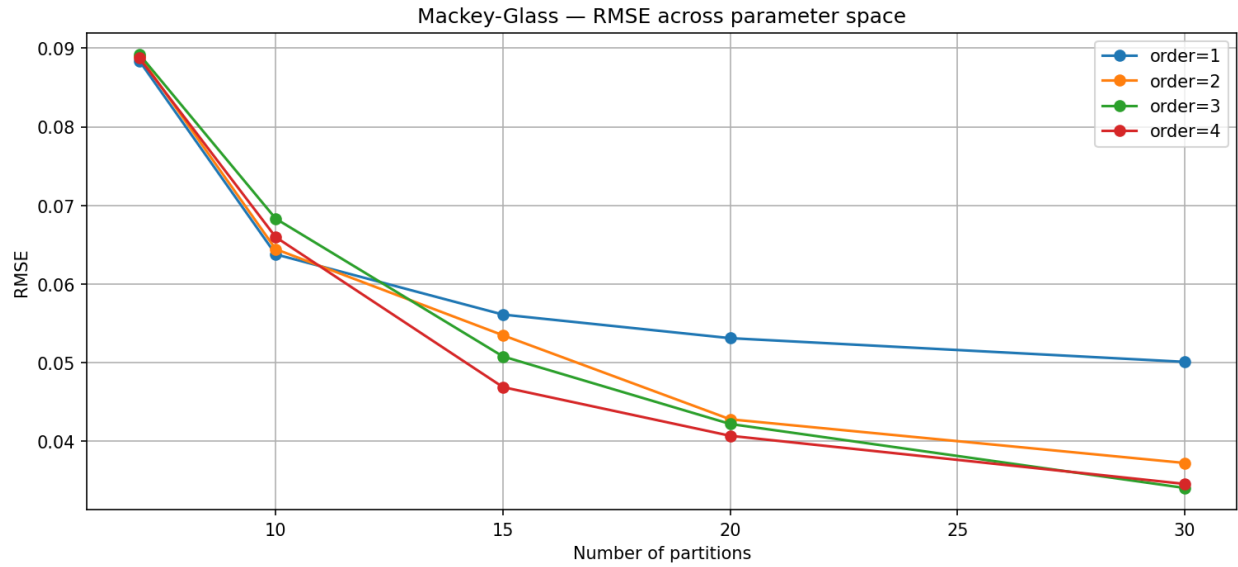


Figure 2. Mackey–Glass: RMSE across parameter space (order $\times$ partitions).

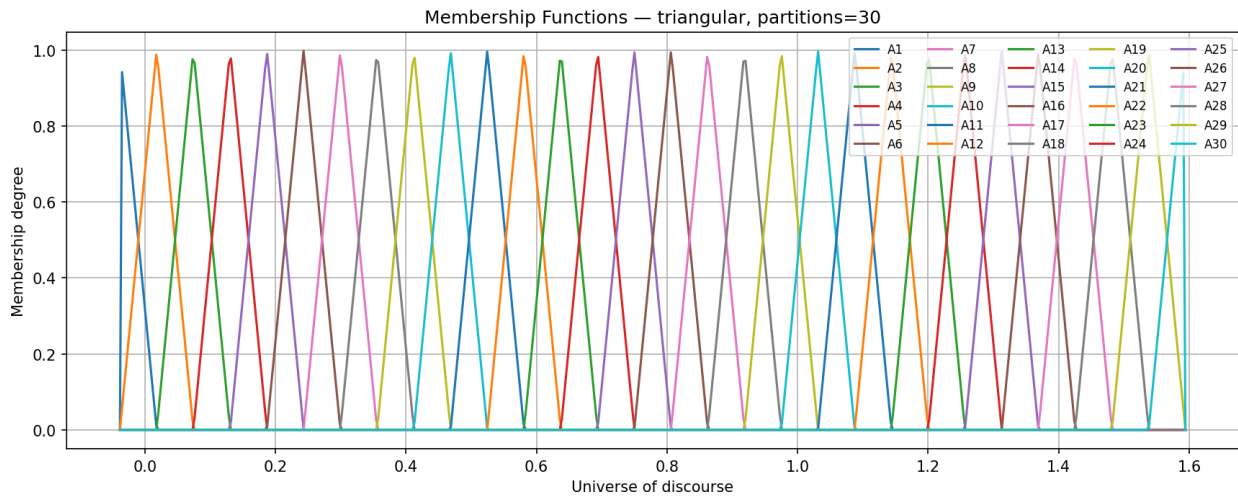


Figure 3. Mackey–Glass: membership functions for the best configuration.

Dataset 2 — Specimens / Influenza

The Influenza dataset includes three forecasting targets: TOTAL SPECIMENS, TOTAL A, and TOTAL B. Because A/B series are typically skewed counts with many small values, experiments apply a $\log_{1p}$ transform during training and evaluate on the original scale after inverse transform and non-negative clamping.

Target: TOTAL SPECIMENS

Model	order	partitions	partitioning	transform	RMSE	MAE	MAPE	n_rules_orderK	hit_rate_any	fallback_rate	backoff_rate
Best (selected by RMSE)	1	15	equal	none	10618	8352.66	10.1309	13	1	0	0
Best FOFTS (order=1)	1	15	equal	none	10618	8352.66	10.1309	13	1	0	0

For this target, the first-order model achieved the lowest RMSE. This suggests that additional lag context does not provide consistent benefits under the tested settings, and higher orders may increase rule sparsity.

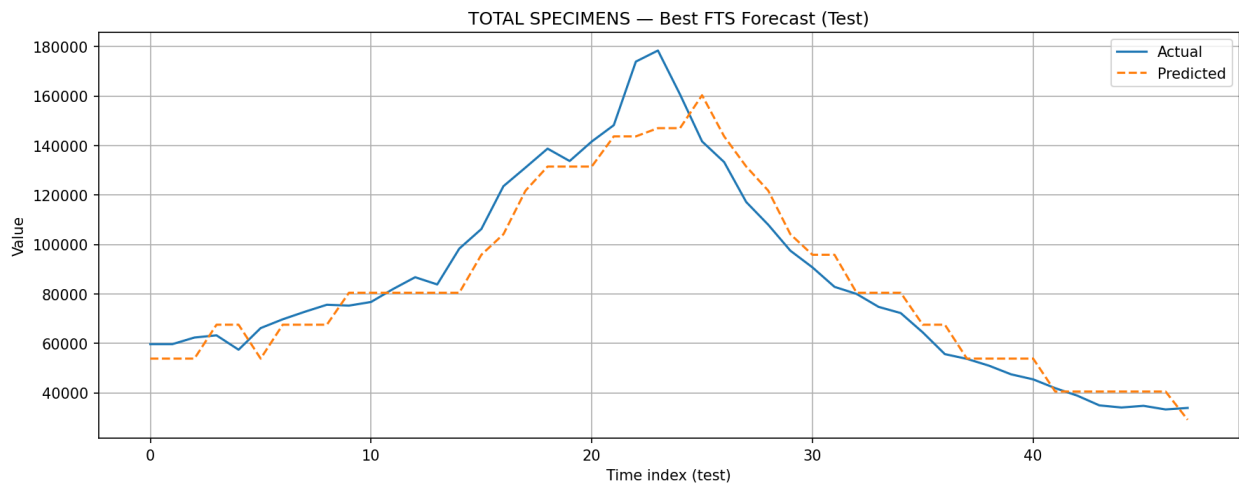


Figure. TOTAL SPECIMENS: actual vs predicted (best configuration, test segment).

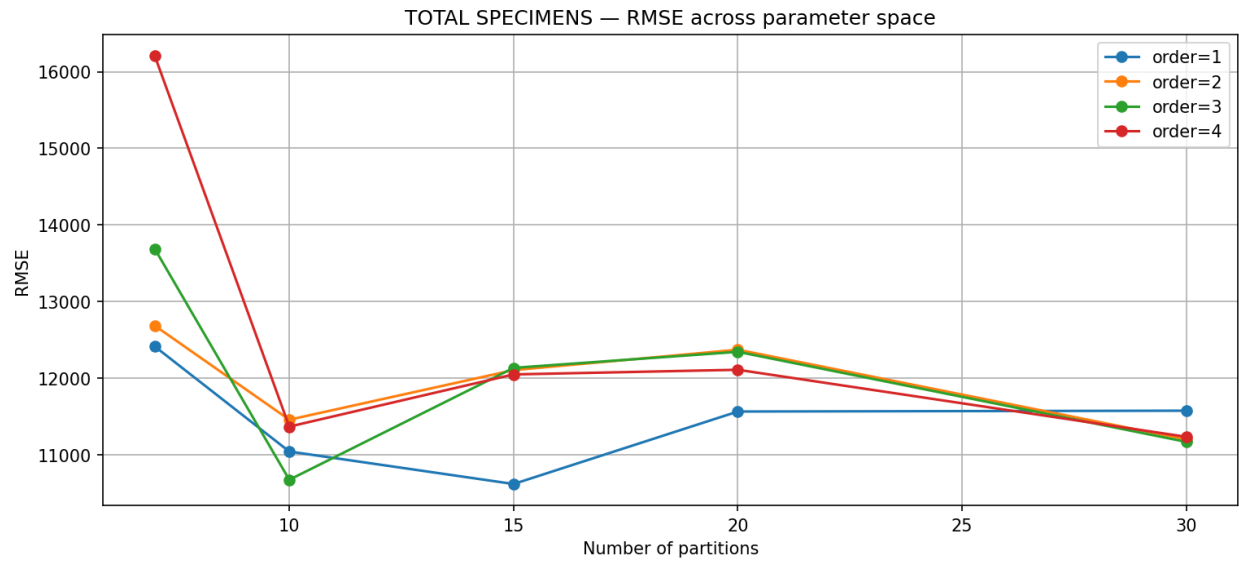


Figure. TOTAL SPECIMENS: RMSE across parameter space (order $\times$ partitions).

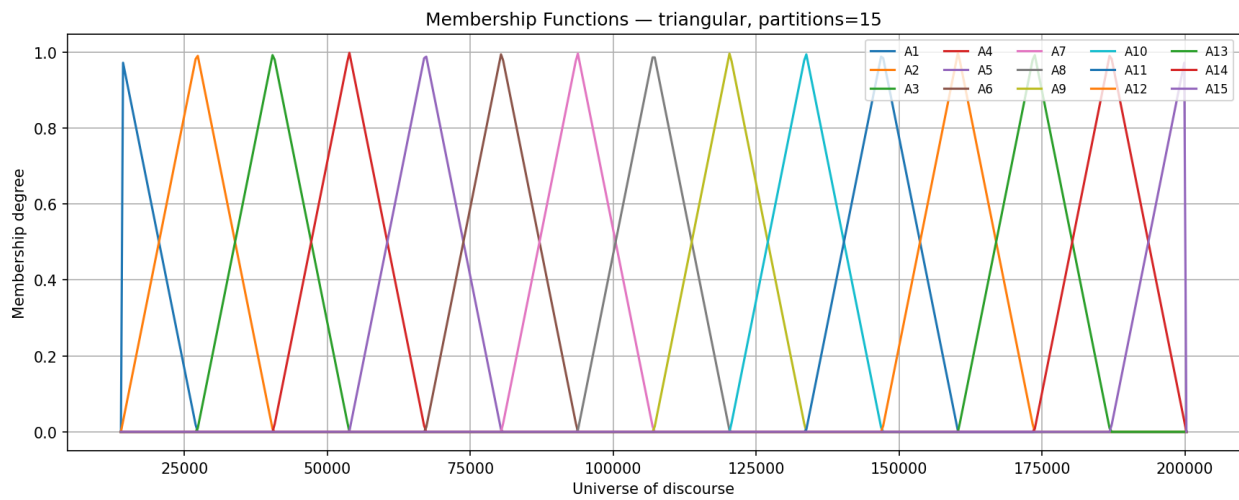


Figure. TOTAL SPECIMENS: membership functions for the best configuration.

Target: TOTAL A

Model	order	partitions	partitioning	transform	RMS E	MA E	MA PE	n_rules_orderK	hit_rate_any	fallback_rate	backoff_rate
Best (selected by RMS E)	3	30	equal	log1p	370 0.41	170 0	25.6 186	124	1	0	0.3541 67
Best FOFTS (order=1)	1	30	equal	log1p	370 1.47	195 5.95	32.0 558	27	1	0	0

For this target, a higher-order model achieved the lowest RMSE. However, the back-off rate indicates frequent use of lower-order rules, highlighting sparsity in exact high-order patterns.

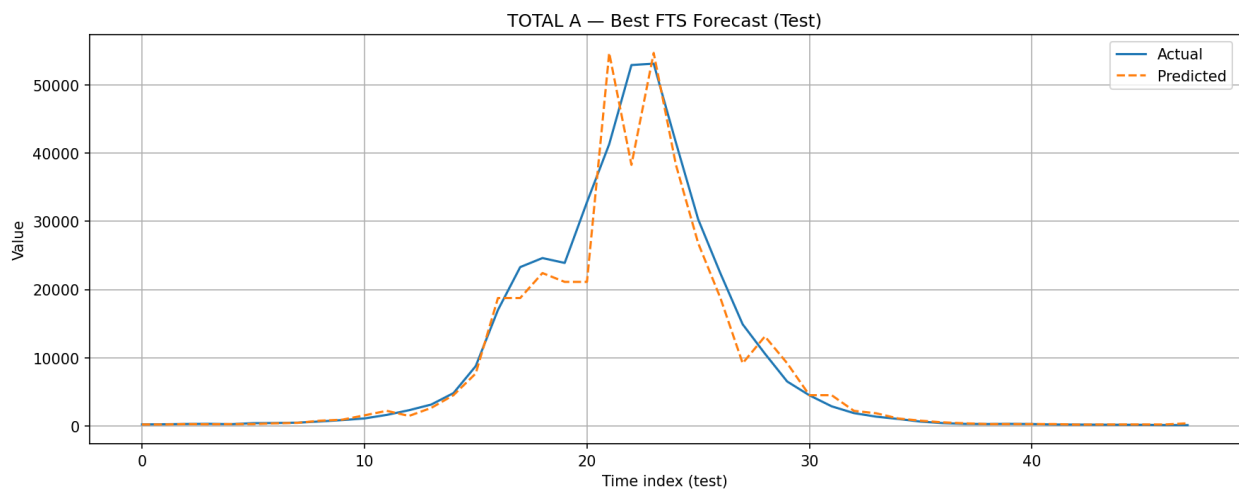


Figure. TOTAL A: actual vs predicted (best configuration, test segment).

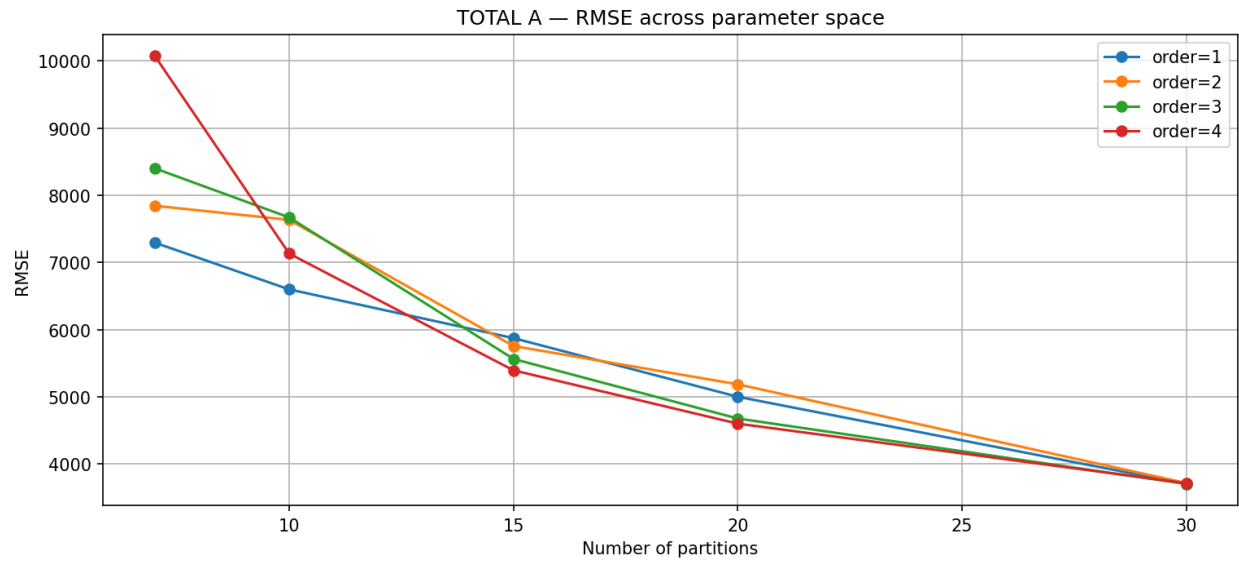


Figure. TOTAL A: RMSE across parameter space (order $\times$ partitions).

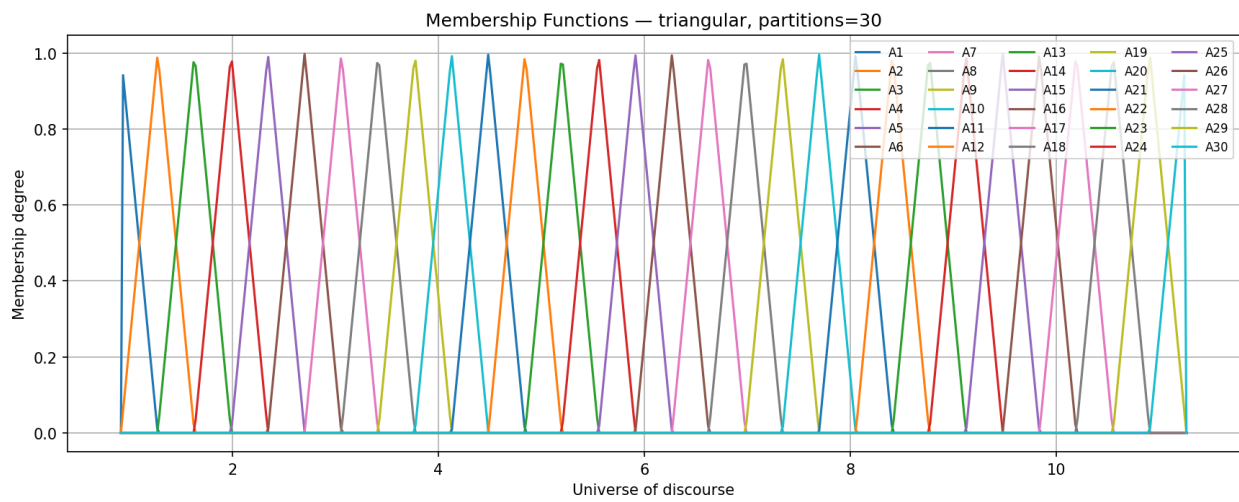


Figure. TOTAL A: membership functions for the best configuration.

Target: TOTAL B

Model	order	partitions	partitioning	transform	RMS E	MA E	MA PE	n_rules_orderK	hit_rate_any	fallback_rate	backoff_rate
Best (selected by RMS E)	1	15	equal	log1p	332.418	214.615	28.3771	13	1	0	0
Best FOFTS (order=1)	1	15	equal	log1p	332.418	214.615	28.3771	13	1	0	0

For this target, the first-order model achieved the lowest RMSE. This suggests that additional lag context does not provide consistent benefits under the tested settings, and higher orders may increase rule sparsity.

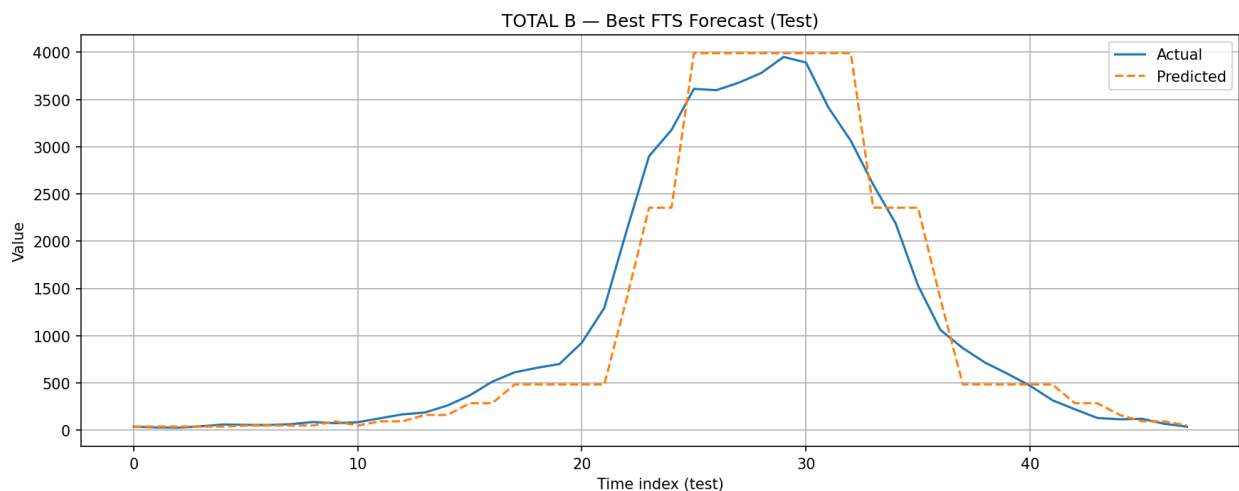


Figure. TOTAL B: actual vs predicted (best configuration, test segment).

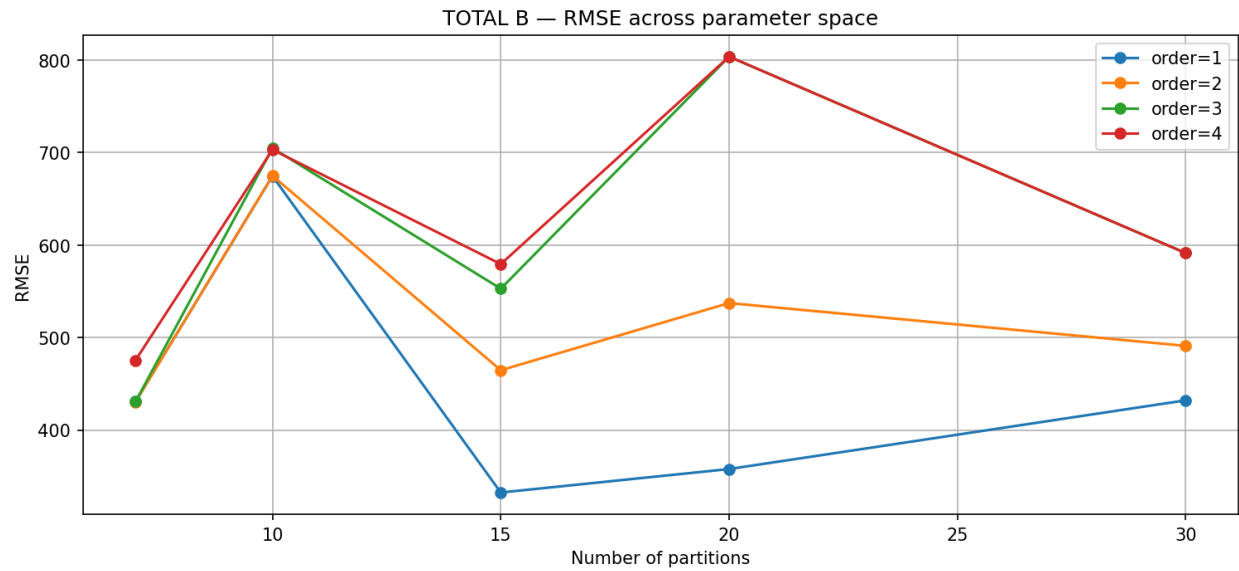


Figure. TOTAL B: RMSE across parameter space (order × partitions).

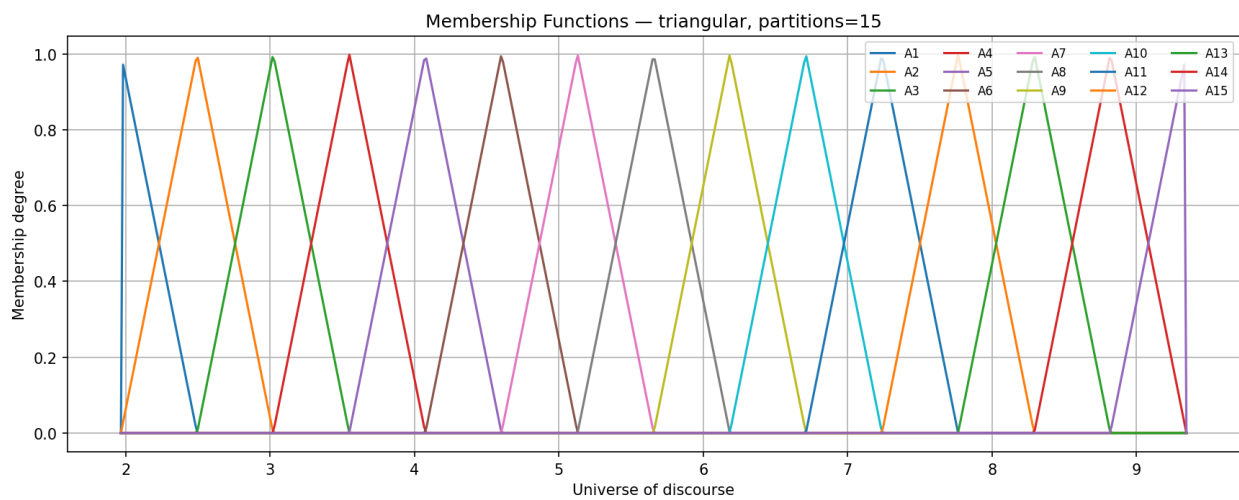


Figure. TOTAL B: membership functions for the best configuration.

FOFTS vs HOFTS Discussion

Across the evaluated datasets, High-Order Fuzzy Time Series (HOFTS) models demonstrate improved predictive performance when temporal dependencies extend beyond a single lag, as clearly observed in the Mackey–Glass benchmark. The chaotic nature of this series benefits from longer historical context, enabling HOFTS to capture more complex dynamical patterns.

However, increasing the model order substantially enlarges the antecedent pattern space, which leads to a rapid growth in rule sparsity and a reduction in exact rule coverage. This phenomenon becomes particularly evident in real-world count data such as the Influenza series, where noise and irregular fluctuations dominate the dynamics.

The implemented true back-off strategy plays a critical role in mitigating this limitation by hierarchically reusing lower-order rules when high-order patterns are unavailable. While this mechanism significantly reduces fallback events, the empirical results indicate that for sparse and noisy datasets (TOTAL A and TOTAL B), the performance gains of HOFTS remain limited and FOFTS often achieves comparable or even superior accuracy.

These findings suggest that the effectiveness of HOFTS is highly data-dependent and that higher-order context is beneficial primarily in systems with strong and structured temporal dependencies.

چکیده فارسی:

در این پروژه، یک چارچوب کامل برای پیش‌بینی سری‌های زمانی فازی (Fuzzy Time Series Forecasting) از صفر و بدون استفاده از کتابخانه‌های آماده پیاده‌سازی و مورد ارزیابی قرار گرفت. هدف اصلی، مقایسه عملکرد مدل‌های فازی مرتبه اول (FOFTS) و مدل‌های فازی مرتبه بالا (HOFTS) در شرایط مختلف و بررسی تاثیر پارامترهای کلیدی مانند تعداد پارتیشن‌ها و مرتبه مدل بر دقت پیش‌بینی بود.

به منظور ارزیابی جامع، دو مجموعه داده متفاوت مورد استفاده قرار گرفت. مجموعه داده اول، سری زمانی Mackey–Glass است و به عنوان معیار استاندارد در مسائل پیش‌بینی شناخته می‌شود. مجموعه داده دوم شامل داده‌های واقعی پایش آنفلوآنزا بوده و سه متغیر TOTAL SPECIMENS، TOTAL A و TOTAL B را در بر می‌گیرد که به ترتیب بیانگر تعداد کل نمونه‌ها، تعداد موارد مبتلا به نوع A و تعداد موارد مبتلا به نوع B هستند.

در مرحله پیش پردازش، داده‌ها به صورت سری زمانی تک متغیره در نظر گرفته شده و بدون درهم‌ریزی، به نسبت ۸۰ درصد برای آموزش و ۲۰ درصد برای آزمون تقسیم شدند تا ساختار زمانی حفظ گردد. فضای گفتمان هر سری زمانی بر اساس کمینه و بیشینه داده‌های آموزشی تعریف شده و به منظور جلوگیری از اثرات مرزی، یک حاشیه پنج درصدی به دو سمت آن افزوده شد.

سپس فضای گفتمان به تعداد مختلفی از مجموعه‌های فازی (7، 10، 15، 20 و 30 بخش) تقسیم گردید و برای هر مجموعه، تابع عضویت مثلثی تعریف شد. فرآیند فازی‌سازی به این صورت انجام گرفت که برای هر مقدار عددی، درجه عضویت آن نسبت به تمام مجموعه‌ها محاسبه و مجموعه‌ای که بیشترین درجه عضویت را داشت به عنوان حالت فازی متناظر انتخاب شد.

در مرحله تولید روابط منطقی فازی، برای مدل مرتبه اول روابط به صورت $A(t-1) \rightarrow A(t)$ و برای مدل‌های مرتبه بالا به صورت $A(t-k), \dots, A(t-1) \rightarrow A(t)$ استخراج گردید. این روابط به صورت ساختارهای داده‌ای مبتنی بر شمارش فراوانی ذخیره شدند تا در فرآیند پیش‌بینی، وزن هر قانون در خروجی لحاظ گردد.

برای تبدیل خروجی فازی به مقدار عددی نهایی، از روش defuzzification وزنی استفاده شد که در آن مراکز مجموعه های فازی پیامد بر اساس فراوانی وزن دهی می شوند. در صورت عدم وجود قانون دقیق، یک استراتژی back-off واقعی پیاده سازی شد که ابتدا به قوانین مرتبه پایین تر مراجعه کرده و در نهایت در صورت نبود هرگونه قانون، مقدار قبلی به عنوان پیش بینی بازگردانده می شود.

نتایج تجربی نشان داد که در داده Mackey-Glass، مدل مرتبه سوم با 30 پارتیشن بهترین عملکرد را داشته و مقدار RMSE آن حدود 32 درصد کمتر از مدل مرتبه اول بوده است. بیانگر آن است که در chaotic systems، استفاده از حافظه زمانی بلند تر موجب استخراج الگوهای پویاتر و بهبود دقت پیش بینی می شود.

در مقابل، در داده های واقعی آنفلوآنزا، مدل های مرتبه بالا بهبود چشمگیری نسبت به مدل مرتبه اول ایجاد نکردند و در برخی موارد، مانند متغیر TOTAL B، مدل مرتبه اول بهترین عملکرد را ارائه داد. این مسئله نشان می دهد که در داده های نویزی و پراکنده، افزایش مرتبه مدل منجر به افزایش کمبود قوانین دقیق و وابستگی بیشتر به سازوکار back-off می شود.

در مجموع، این پروژه نشان می دهد که مدل های FTS ابزاری مناسب برای پیش بینی سری های زمانی غیرخطی هستند و مزیت اصلی آنها در تفسیرپذیری و ساختار قانون محور نهفته است. با این حال، انتخاب مرتبه مناسب مدل به ماهیت داده وابسته بوده و استفاده از مدل های مرتبه بالا تنها در شرایط وجود وابستگی زمانی قوی منجر به بهبود معنادار عملکرد می شود.